Series 3 Survival Analysis

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(I) Log-likelihood for Lognormal regression with right censoring

We assume an Accelerated Failure Time (AFT) lognormal regression model:

$$\log T_i = \mu_i + \sigma\varepsilon_i, \quad \varepsilon_i \sim N(0,1),$$

where

$$\mu_i = x_i^\top \beta,$$

and:

- T_i: survival time
- x_i: covariate vector
- β: regression coefficients
- $\sigma > 0$: scale parameter
- $\delta_i = 1$: event observed
- $\delta_i = 0$: right-censored observation

Since $\log T_i \sim N(\mu_i, \sigma^2)$, the density of T_i given x_i is:

$$f(t_i \mid x_i) = \frac{1}{t_i \sigma \sqrt{2\pi}} \exp\left(-\frac{(\log t_i - \mu_i)^2}{2\sigma^2}\right), \quad t_i > 0.$$

The survival function is:

$$S(t_i \mid x_i) = 1 - \Phi\left(\frac{\log t_i - \mu_i}{\sigma}\right),$$

where Φ is the standard normal CDF. Define:

$$z_i = \frac{\log t_i - \mu_i}{\sigma} = \frac{\log t_i - x_i^\top \beta}{\sigma}.$$

For each observation:

- if $\delta_i = 1$, the contribution is $f(t_i \mid x_i)$,
- if $\delta_i = 0$, the contribution is $S(t_i \mid x_i)$.

Thus, the likelihood is:

$$L(\beta, \sigma) = \prod_{i=1}^{n} [f(t_i \mid x_i)]^{\delta_i} [S(t_i \mid x_i)]^{1-\delta_i}.$$

Taking logs, the log-likelihood becomes:

$$\ell(\beta, \sigma) = \sum_{i=1}^{n} \{\delta_i \log f(t_i \mid x_i) + (1 - \delta_i) \log S(t_i \mid x_i)\}.$$

We expand the terms:

$$\log f(t_i \mid x_i) = -\log t_i - \log \sigma - \tfrac{1}{2} \log(2\pi) - \frac{(\log t_i - x_i^\top \beta)^2}{2\sigma^2},$$

$$\log S(t_i \mid x_i) = \log \left[1 - \Phi\left(\frac{\log t_i - x_i^\top \beta}{\sigma}\right)\right].$$

Therefore, the full log-likelihood is:

$$\ell(\beta, \sigma) = \sum_{i=1}^{n} \delta_i \left[-\log t_i - \log \sigma - \tfrac{1}{2} \log(2\pi) - \frac{(\log t_i - x_i^\top \beta)^2}{2\sigma^2}\right] + \sum_{i=1}^{n} (1-\delta_i) \log \left[1 - \Phi\left(\frac{\log t_i - x_i^\top \beta}{\sigma}\right)\right].$$

(II)

In this part of the assignment we are going to analyze a lung-cancer dataset whichcontains survival information for lung cancer patients, along with several clinical covariates.

Survival variables

- **time**: Survival time in days

- **status**: Event indicator (`1` = event observed (death) ,`0` = right-censored observation)

Covariates

- **age**: Age of the patient (in years)

- **treatment**: Type of treatment (`0` = Treatment type 1 , `1` = Treatment type 2)

- **cell.type**: Histological type of lung cancer (1 = Squamous cell carcinoma, 2 = Small cell carcinoma, 3 = Adenocarcinoma , 4 = Large cell carcinoma)
- **karno.score**: Karnofsky performance score (0 = poor condition, 100 = excellent condition)
- **months**: Number of months since diagnosis
- **prior.therapy**: Whether the patient received prior therapy (0 = No prior therapy, 1 = Yes, prior therapy was given)

```
data <- read.table("/home/kontadak/Documents/File Hub/Semfe Grad/Survival Analysis/Series 3/l
                   header = TRUE)

# Inspect structure
head(data)
```

```
  id cell.type time status treatment karno.score months age prior.therapy
1  1         1   72      1         1          60      7  69             0
2  2         1  411      0         1          70      5  64             1
3  3         1  228      1         1          60      3  38             0
4  4         1  126      1         1          60      9  63             1
5  5         1  118      1         1          70     11  65             1
6  6         1   10      1         1          20      5  49             0
```

```
str(data)
```

```
'data.frame':   137 obs. of  9 variables:
 $ id           : int  1 2 3 4 5 6 7 8 9 10 ...
 $ cell.type    : int  1 1 1 1 1 1 1 1 1 1 ...
 $ time         : int  72 411 228 126 118 10 82 110 314 100 ...
 $ status       : int  1 0 1 1 1 1 1 1 1 0 ...
 $ treatment    : int  1 1 1 1 1 1 1 1 1 1 ...
 $ karno.score  : int  60 70 60 60 70 20 40 80 50 70 ...
 $ months       : int  7 5 3 9 11 5 10 29 18 6 ...
 $ age          : int  69 64 38 63 65 49 69 68 43 70 ...
 $ prior.therapy: int  0 1 0 1 1 0 1 0 0 0 ...
```

```
# Convert cell.type to factor with labels
data$cell.type <- factor(data$cell.type,
                         levels = c(1,2,3,4),
                         labels = c("Squamous", "Small cell", "Adeno", "Large"))
```

```r
# Summary by cell type
cat("Sample size by cell type:\n")
```

```
Sample size by cell type:
```

```r
print(table(data$cell.type))
```

```

  Squamous Small cell      Adeno      Large 
        35         48         27         27 
```

```r
cat("\nCensoring status by cell type:\n")
```

```

Censoring status by cell type:
```

```r
print(table(data$cell.type, data$status))
```

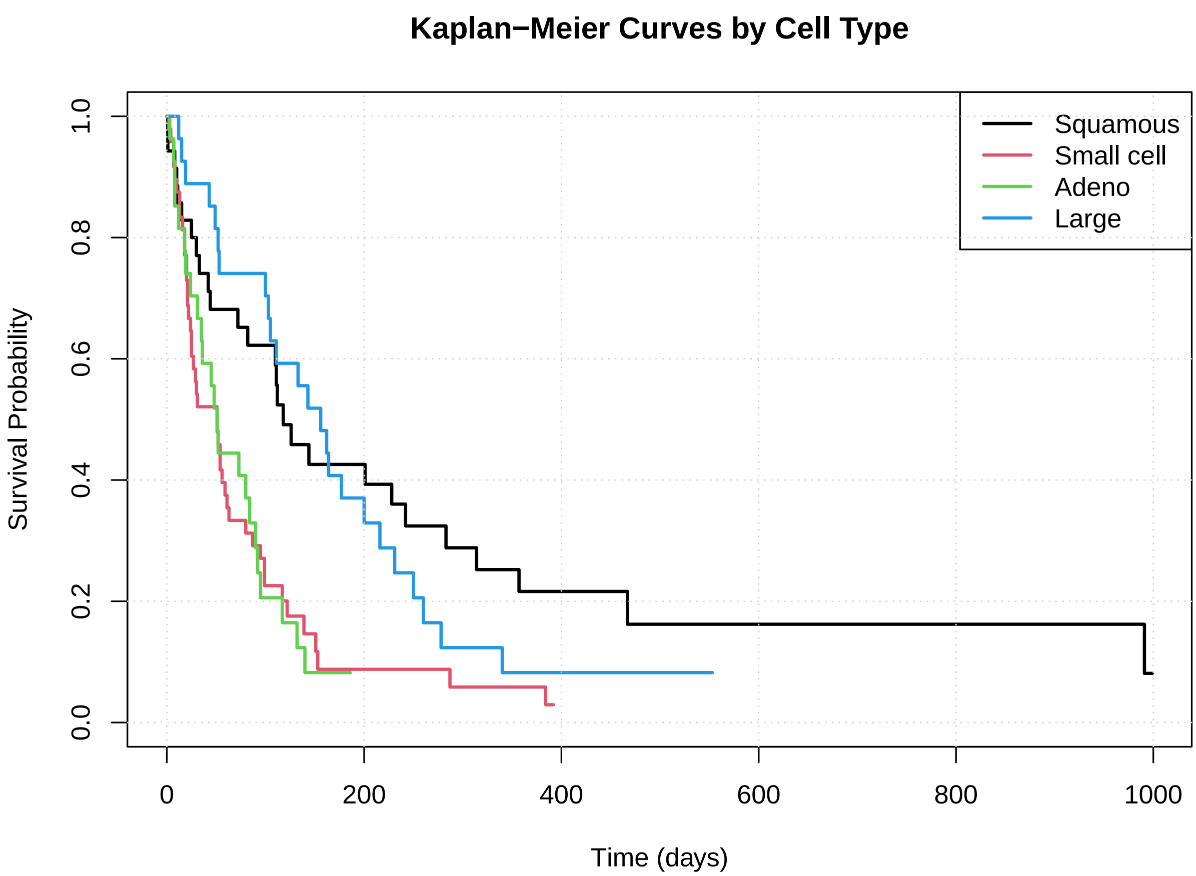
```

              0  1
  Squamous    8 27
  Small cell  4 44
  Adeno       3 24
  Large       3 24
```

A)

For the covariate cell.type we will do log-rank and Wilconox diagnostics as plot the Kaplan-Meier.

```r
library(survival)

fit_km <- survfit(Surv(time, status) ~ cell.type, data = data)

plot(fit_km, col = 1:4, lwd = 2,
     xlab = "Time (days)", ylab = "Survival Probability",
     main = "Kaplan-Meier Curves by Cell Type")
```

```r
legend("topright", legend = levels(data$cell.type),
       col = 1:4, lwd = 2)
grid()
```




```r
# Log-rank test (rho = 0)
logrank_test <- survdiff(Surv(time, status) ~ cell.type, data = data)
logrank_test
```

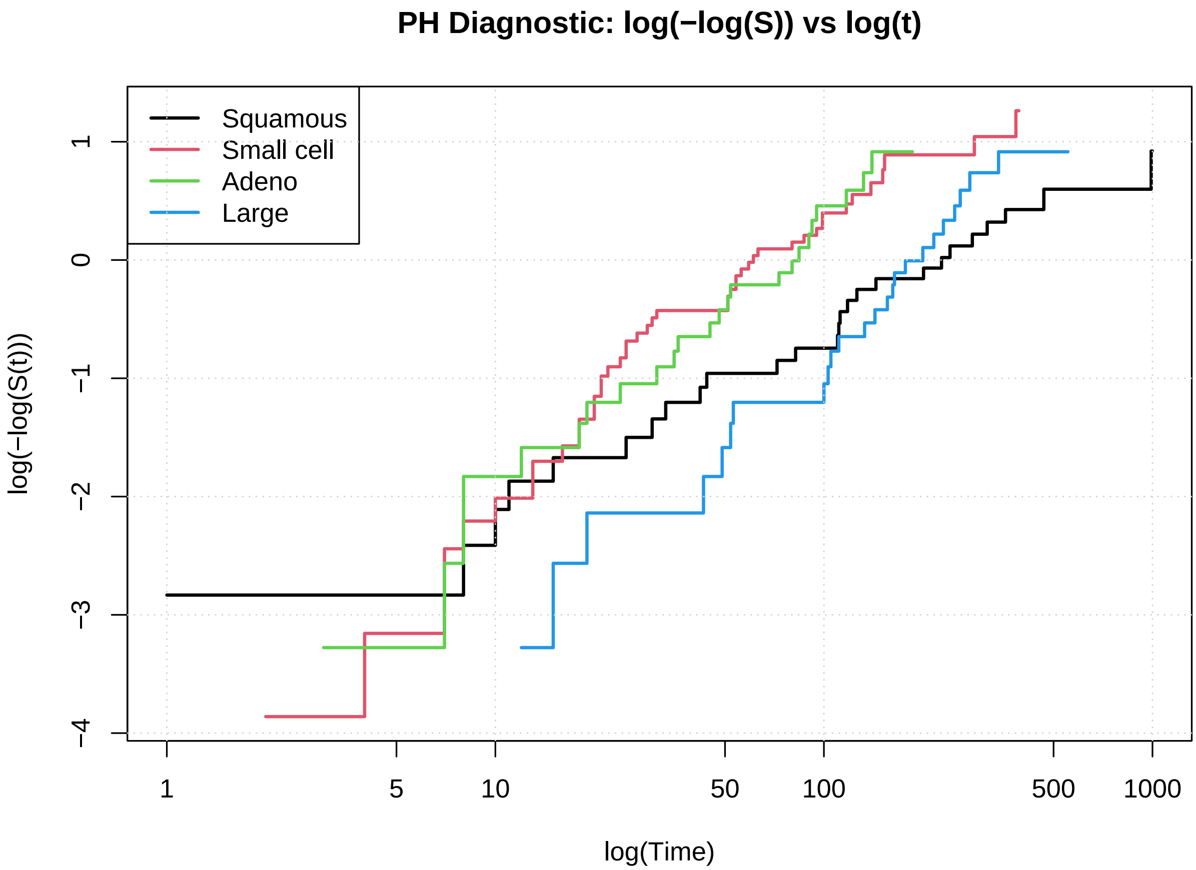
```
Call:
survdiff(formula = Surv(time, status) ~ cell.type, data = data)

                        N Observed Expected (O-E)^2/E (O-E)^2/V
cell.type=Squamous   35       27     41.1      4.83      7.97
cell.type=Small cell 48       44     29.4      7.22     10.03
cell.type=Adeno      27       24     15.6      4.54      5.49
cell.type=Large      27       24     32.9      2.41      3.44
```

```
 Chisq= 21.1  on 3 degrees of freedom, p= 1e-04
```

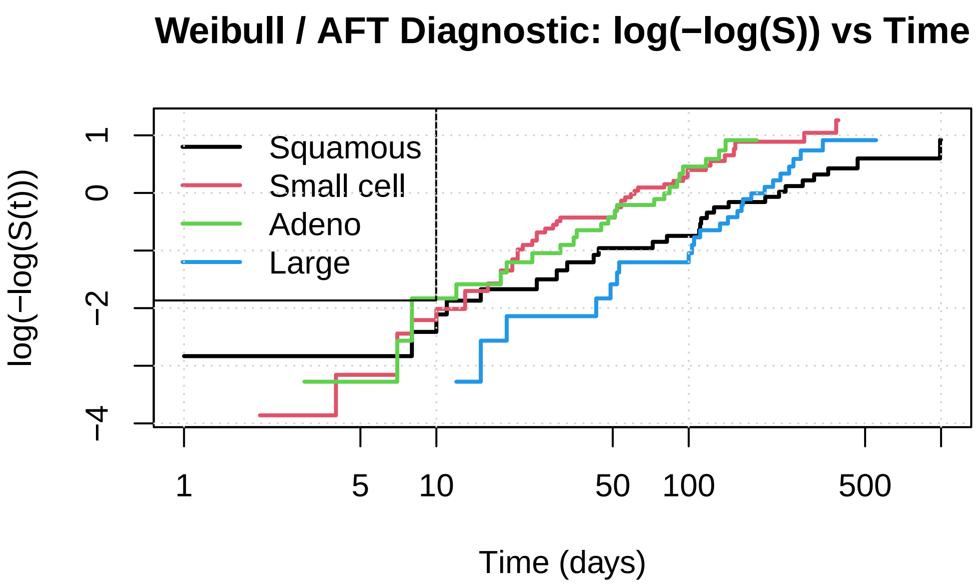
```r
# Wilcoxon test (rho = 1)
wilcox_test <- survdiff(Surv(time, status) ~ cell.type, data = data, rho = 1)
wilcox_test
```

```
Call:
survdiff(formula = Surv(time, status) ~ cell.type, data = data,
    rho = 1)

                      N Observed Expected (O-E)^2/E (O-E)^2/V
cell.type=Squamous   35    13.39     19.9      2.11      4.64
cell.type=Small cell 48    28.39     19.0      4.70      9.36
cell.type=Adeno      27    15.59     10.8      2.14      3.53
cell.type=Large      27     9.54     17.3      3.49      7.30

 Chisq= 18.6  on 3 degrees of freedom, p= 3e-04
```

```r
plot(fit_km, fun = "cloglog", col = 1:4, lwd = 2,
     xlab = "log(Time)", ylab = "log(-log(S(t)))",
     main = "PH Diagnostic: log(-log(S)) vs log(t)")

legend("topleft", legend = levels(data$cell.type),
       col = 1:4, lwd = 2)
grid()
```




```
plot(fit_km, fun = "cloglog", col = 1:4, lwd = 2,
     xscale = 1, # ensures time is on x-axis
     xlab = "Time (days)", ylab = "log(-log(S(t)))",
     main = "Weibull / AFT Diagnostic: log(-log(S)) vs Time")

legend("topleft", legend = levels(data$cell.type),
       col = 1:4, lwd = 2)
grid()
```



The Kaplan–Meier curves show clear visual differences in survival among the four histological cell types. Small-cell carcinoma and adeno exhibit the poorest survival, while squamous and large-cell types show comparatively better survival patterns.

The log-rank test strongly rejects the null hypothesis of equal survival functions across the four groups ($\chi^2 = 21.1, df = 3, p = 10^{-4}$). The Wilcoxon test, which places more weight on early failures, also indicates significant differences ($\chi^2 = 18.6, df = 3, p = 3 \cdot 10^{-4}$). Both tests confirm that survival differs significantly by cell type.

The proportional hazards (PH) diagnostic plot, based on log(−log(S(t))) versus log(time), shows curves that are not parallel across the four groups, suggesting that the PH assumption may not hold well for this covariate. The AFT diagnostic plot (log(time) vs. S(t)) also shows noticeable divergence between groups, indicating that the AFT assumption is not perfectly satisfied either.

Overall, the graphical diagnostics suggest that neither the PH nor the AFT assumptions hold cleanly for the cell type variable when considered alone. However, the Weibull model which is compatible with both PH and AFT formulations—may still provide a reasonable parametric fit and will be examined later.

(B)

i)

```r
library(survival)

# Full Weibull AFT model
fit_w_full <- survreg(Surv(time, status) ~ age + treatment + cell.type +
                        karno.score + months + prior.therapy,
                      data = data,
                      dist = "weibull")

summary(fit_w_full)      # Wald tests (z, p-values)
```

```

Call:
survreg(formula = Surv(time, status) ~ age + treatment + cell.type +
    karno.score + months + prior.therapy, data = data, dist = "weibull")
                      Value Std. Error     z       p
(Intercept)         3.18737    0.75046  4.25 2.2e-05
age                 0.00952    0.00923  1.03  0.3024
treatment          -0.23811    0.20379 -1.17  0.2427
cell.typeSmall cell -0.91563    0.27021 -3.39  0.0007
cell.typeAdeno     -1.16899    0.28778 -4.06 4.9e-05
cell.typeLarge     -0.45047    0.28340 -1.59  0.1119
karno.score         0.03418    0.00526  6.50 7.9e-11
months             -0.00074    0.00860 -0.09  0.9314
prior.therapy      -0.09229    0.23167 -0.40  0.6903
Log(scale)         -0.01388    0.06989 -0.20  0.8426

Scale= 0.986

Weibull distribution
Loglik(model)= -667.9   Loglik(intercept only)= -701.1
    Chisq= 66.44 on 8 degrees of freedom, p= 2.5e-11
Number of Newton-Raphson Iterations: 5
n= 137
```

```r
AIC(fit_w_full)          # AIC
```

```
[1] 1355.767
```

```r
# Example: drop 'months'
fit_w_no_months <- update(fit_w_full, . ~ . - months)
```

```r
anova(fit_w_no_months, fit_w_full, test = "Chisq")
```

```
                                                                Terms Resid. Df
1          age + treatment + cell.type + karno.score + prior.therapy       128
2 age + treatment + cell.type + karno.score + months + prior.therapy       127
    -2*LL    Test Df    Deviance Pr(>Chi)
1 1335.774         NA          NA       NA
2 1335.767 +months  1 0.007340812 0.931722
```

```r
library(MASS)

fit_w_step <- stepAIC(fit_w_full, direction = "backward", trace = TRUE)
```

```
Start:  AIC=1355.77
Surv(time, status) ~ age + treatment + cell.type + karno.score +
    months + prior.therapy

                Df    AIC
- months         1 1353.8
- prior.therapy  1 1353.9
- age            1 1354.8
- treatment      1 1355.1
<none>             1355.8
- cell.type      3 1368.3
- karno.score    1 1391.8

Step:  AIC=1353.77
Surv(time, status) ~ age + treatment + cell.type + karno.score +
    prior.therapy

                Df    AIC
- prior.therapy  1 1352.0
- age            1 1352.8
- treatment      1 1353.2
<none>             1353.8
- cell.type      3 1366.4
- karno.score    1 1391.1

Step:  AIC=1352.01
Surv(time, status) ~ age + treatment + cell.type + karno.score
```

```
               Df    AIC
- age           1 1351.2
- treatment     1 1351.6
<none>            1352.0
- cell.type     3 1364.5
- karno.score   1 1389.2

Step:  AIC=1351.15
Surv(time, status) ~ treatment + cell.type + karno.score

               Df    AIC
- treatment     1 1350.3
<none>            1351.2
- cell.type     3 1362.7
- karno.score   1 1387.3

Step:  AIC=1350.29
Surv(time, status) ~ cell.type + karno.score

               Df    AIC
<none>            1350.3
- cell.type     3 1361.0
- karno.score   1 1386.0
```

```r
summary(fit_w_step)
```

```
Call:
survreg(formula = Surv(time, status) ~ cell.type + karno.score,
    data = data, dist = "weibull")
                        Value Std. Error     z       p
(Intercept)          3.362848   0.370124  9.09 < 2e-16
cell.typeSmall cell -0.787908   0.251501 -3.13  0.0017
cell.typeAdeno      -1.125254   0.284017 -3.96 7.4e-05
cell.typeLarge      -0.379734   0.280929 -1.35  0.1765
karno.score          0.032879   0.005063  6.49 8.4e-11
Log(scale)          -0.000731   0.069476 -0.01  0.9916

Scale= 0.999

Weibull distribution
```

```
Loglik(model)= -669.1   Loglik(intercept only)= -701.1
    Chisq= 63.92 on 4 degrees of freedom, p= 4.4e-13
Number of Newton-Raphson Iterations: 5
n= 137
```

```r
AIC(fit_w_step)
```

```
[1] 1350.293
```

A Weibull AFT regression model was fitted using all available covariates. Wald tests from the full model indicated that only the histological cell type and the Karnofsky performance score were statistically significant predictors of survival, while age, treatment type, months since diagnosis, and prior therapy showed no meaningful contribution.

Likelihood ratio tests comparing nested models confirmed that removing non-significant variables (e.g., months, prior.therapy, age, treatment) did not significantly reduce model fit.

Backward stepwise selection using AIC led to a final model including only **cell.type** and **karno.score**, achieving the lowest AIC value:

$$\mathrm{AIC}_{\text{final}} = 1350.29.$$

The final Weibull model remained highly significant overall, with likelihood ratio statistic

$$\chi^2 = 63.92, \qquad df = 4, \qquad p = 4.4 \times 10^{-13}.$$

Thus, the best-fitting Weibull model for these data includes only the histological cell type and the Karnofsky performance score.

ii) We are now going to create 95% confidence intervals for the covariate's parameters.

```r
# Final Weibull AFT model
fit_w_step <- survreg(Surv(time, status) ~ cell.type + karno.score,
                      data = data, dist = "weibull")

# Extract coefficients (excluding the scale parameter)
beta_hat <- coef(fit_w_step)[names(coef(fit_w_step)) != "Log(scale)"]

# Extract standard errors (same exclusion)
se_hat <- sqrt(diag(vcov(fit_w_step)))[names(beta_hat)]

# 95% Wald confidence intervals for coefficients
```

```r
lower_beta <- beta_hat - 1.96 * se_hat
upper_beta <- beta_hat + 1.96 * se_hat

CI_beta <- cbind(Estimate = beta_hat,
                 Lower95 = lower_beta,
                 Upper95 = upper_beta)

CI_beta
```

```
                    Estimate     Lower95     Upper95
(Intercept)       3.36284835  2.63740563  4.08829106
cell.typeSmall cell -0.78790843 -1.28084974 -0.29496712
cell.typeAdeno    -1.12525392 -1.68192775 -0.56858008
cell.typeLarge    -0.37973391 -0.93035448  0.17088666
karno.score        0.03287872  0.02295519  0.04280225
```

```r
# ---- Time ratios (AFT interpretation) ----
TR       <- exp(beta_hat)
lower_TR <- exp(lower_beta)
upper_TR <- exp(upper_beta)

CI_TR <- cbind(TimeRatio = TR,
               Lower95 = lower_TR,
               Upper95 = upper_TR)

CI_TR
```

```
                    TimeRatio    Lower95    Upper95
(Intercept)        28.8713094 13.9768953 59.6378870
cell.typeSmall cell  0.4547950  0.2778011  0.7445561
cell.typeAdeno      0.3245700  0.1860150  0.5663290
cell.typeLarge      0.6840434  0.3944139  1.1863563
karno.score         1.0334252  1.0232207  1.0437315
```

The results show the following:

- Small cell carcinoma has a time ratio of approximately 0.45, with a 95% CI well below 1. This indicates that patients with small-cell carcinoma have about 55% shorter expected survival time compared to the Squamous baseline.

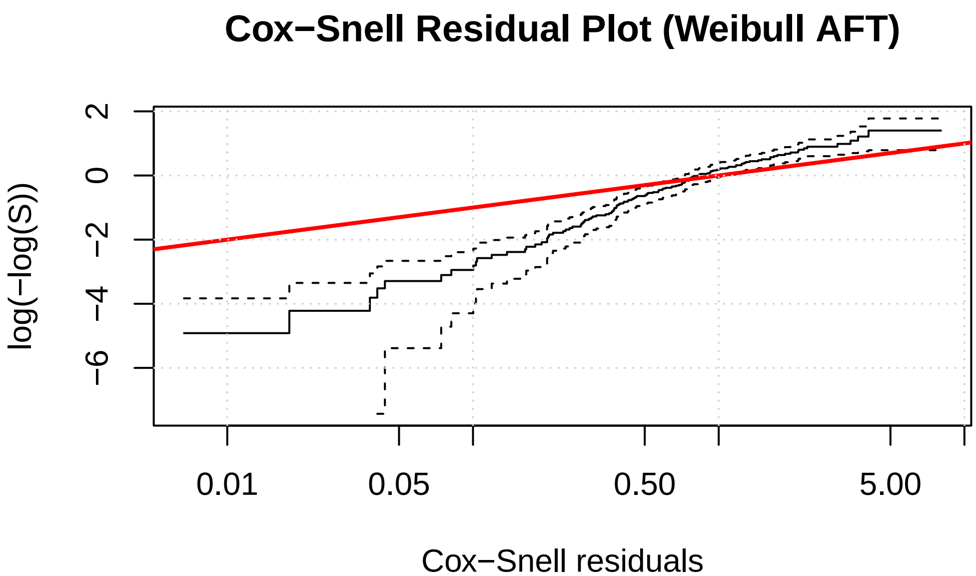
- Adenocarcinoma has an even smaller time ratio (around 0.32), with a 95% CI also entirely below 1. This suggests substantially shorter survival, making it the poorest-prognosis group relative to Squamous.
- Large cell carcinoma has a time ratio near 0.68, but its 95% CI includes 1. Thus, there is no statistically significant difference in expected survival compared to Squamous.
- Karnofsky performance score has a time ratio of approximately 1.033, with a 95% CI entirely above 1. This means that each 1-point increase in Karnofsky score is associated with a 3.3% increase in expected survival time, a strong and statistically significant effect.

Overall, the confidence intervals confirm that histological cell type and Karnofsky performance score are the key predictors of survival in the Weibull AFT model, with poorer-prognosis cell types showing markedly reduced survival times and higher performance scores associated with longer survival.

iii)

```
library(survival)

# Final Weibull AFT model
fit_w_step <- survreg(Surv(time, status) ~ cell.type + karno.score,
                      data = data, dist = "weibull")

# Extract Weibull parameters
sigma <- fit_w_step$scale
shape <- 1 / sigma   # Weibull shape parameter

# Linear predictor
eta <- predict(fit_w_step, type = "lp")

# Cox-Snell residuals
H_hat <- (data$time / exp(eta))^shape

# KM estimate of Cox-Snell residuals
fit_cs <- survfit(Surv(H_hat, data$status) ~ 1)

# Plot: KM vs Exp(1)
plot(
  fit_cs,
  fun = "cloglog",
  xlab = "Cox-Snell residuals",
  ylab = "log(-log(S))",
  main = "Cox-Snell Residual Plot (Weibull AFT)"
```

```r
)

abline(0, 1, col = "red", lwd = 2)
grid()
```



The Cox–Snell residual plot shows that the empirical curve lies close to the theoretical Exp(1) line, indicating that the Weibull AFT model fits the data well, with only mild deviations in the extreme tail.

(C)

i)

```r
library(survival)

# Full Cox model
fit_cox_full <- coxph(
  Surv(time, status) ~ cell.type + karno.score +
    age + treatment + months + prior.therapy,
  data = data
)

summary(fit_cox_full)     # Wald tests, HRs, p-values
```

```
Call:
coxph(formula = Surv(time, status) ~ cell.type + karno.score +
    age + treatment + months + prior.therapy, data = data)

  n= 137, number of events= 119

                        coef exp(coef)  se(coef)      z Pr(>|z|)
cell.typeSmall cell  0.836841  2.309060  0.276135  3.031 0.002441 **
cell.typeAdeno       1.069025  2.912538  0.307493  3.477 0.000508 ***
cell.typeLarge       0.338116  1.402303  0.290631  1.163 0.244672
karno.score         -0.035320  0.965296  0.005627 -6.276 3.46e-10 ***
age                 -0.012396  0.987680  0.009338 -1.328 0.184339
treatment            0.332791  1.394856  0.211013  1.577 0.114770
months              -0.001312  0.998689  0.008973 -0.146 0.883764
prior.therapy        0.152360  1.164579  0.236308  0.645 0.519089
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                    exp(coef) exp(-coef) lower .95 upper .95
cell.typeSmall cell    2.3091     0.4331    1.3440     3.967
cell.typeAdeno         2.9125     0.3433    1.5942     5.321
cell.typeLarge         1.4023     0.7131    0.7933     2.479
karno.score            0.9653     1.0360    0.9547     0.976
age                    0.9877     1.0125    0.9698     1.006
treatment              1.3949     0.7169    0.9224     2.109
months                 0.9987     1.0013    0.9813     1.016
prior.therapy          1.1646     0.8587    0.7329     1.851

Concordance= 0.741  (se = 0.021 )
Likelihood ratio test= 62.03  on 8 df,   p=2e-10
Wald test            = 61.47  on 8 df,   p=2e-10
Score (logrank) test = 66.11  on 8 df,   p=3e-11
```

```r
anova(fit_cox_full, test = "LRT")   # Likelihood ratio test
```

```
Analysis of Deviance Table
 Cox model: response is Surv(time, status)
Terms added sequentially (first to last)

               loglik   Chisq Df
NULL          -488.19
cell.type     -477.82 20.7265  3
```

```
karno.score   -459.29 37.0684  1
age           -458.75  1.0676  1
treatment     -457.39  2.7371  1
months        -457.37  0.0239  1
prior.therapy -457.17  0.4095  1
```

```r
AIC(fit_cox_full)         # AIC
```

```
[1] 930.3373
```

```r
# Drop months
fit_cox_1 <- update(fit_cox_full, . ~ . - months)
anova(fit_cox_full, fit_cox_1, test = "LRT"); AIC(fit_cox_1)
```

```
Analysis of Deviance Table
 Cox model: response is  Surv(time, status)
 Model 1: ~ cell.type + karno.score + age + treatment + months + prior.therapy
 Model 2: ~ cell.type + karno.score + age + treatment + prior.therapy
   loglik  Chisq Df Pr(>|Chi|)
1 -457.17
2 -457.18 0.0217  1     0.8828
```

```
[1] 928.359
```

```r
# Drop prior.therapy
fit_cox_2 <- update(fit_cox_1, . ~ . - prior.therapy)
anova(fit_cox_1, fit_cox_2, test = "LRT"); AIC(fit_cox_2)
```

```
Analysis of Deviance Table
 Cox model: response is  Surv(time, status)
 Model 1: ~ cell.type + karno.score + age + treatment + prior.therapy
 Model 2: ~ cell.type + karno.score + age + treatment
   loglik  Chisq Df Pr(>|Chi|)
1 -457.18
2 -457.39 0.4116  1     0.5211
```

```
[1] 926.7707
```

```r
# Drop age
fit_cox_3 <- update(fit_cox_2, . ~ . - age)
anova(fit_cox_2, fit_cox_3, test = "LRT"); AIC(fit_cox_3)
```

```
Analysis of Deviance Table
 Cox model: response is  Surv(time, status)
 Model 1: ~ cell.type + karno.score + age + treatment
 Model 2: ~ cell.type + karno.score + treatment
   loglik  Chisq Df Pr(>|Chi|)
1 -457.39
2 -458.29 1.8049  1     0.1791
```

```
[1] 926.5756
```

```r
# Drop treatment
fit_cox_4 <- update(fit_cox_3, . ~ . - treatment)
anova(fit_cox_3, fit_cox_4, test = "LRT"); AIC(fit_cox_4)
```

```
Analysis of Deviance Table
 Cox model: response is  Surv(time, status)
 Model 1: ~ cell.type + karno.score + treatment
 Model 2: ~ cell.type + karno.score
   loglik  Chisq Df Pr(>|Chi|)
1 -458.29
2 -459.29 1.9999  1     0.1573
```

```
[1] 926.5754
```

```r
# Final model
fit_cox_final <- fit_cox_4
summary(fit_cox_final)
```

```
Call:
coxph(formula = Surv(time, status) ~ cell.type + karno.score,
    data = data)

  n= 137, number of events= 119

                         coef exp(coef)  se(coef)      z Pr(>|z|)
```

```
cell.typeSmall cell  0.669118  1.952515  0.255594  2.618 0.008847 **
cell.typeAdeno       1.017054  2.765037  0.299580  3.395 0.000686 ***
cell.typeLarge       0.270330  1.310396  0.286671  0.943 0.345683
karno.score         -0.032797  0.967735  0.005279 -6.213 5.21e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                    exp(coef) exp(-coef) lower .95 upper .95
cell.typeSmall cell    1.9525     0.5122    1.1831    3.2222
cell.typeAdeno         2.7650     0.3617    1.5371    4.9740
cell.typeLarge         1.3104     0.7631    0.7471    2.2984
karno.score            0.9677     1.0333    0.9578    0.9778

Concordance= 0.737  (se = 0.022 )
Likelihood ratio test= 57.79  on 4 df,   p=8e-12
Wald test            = 59.78  on 4 df,   p=3e-12
Score (logrank) test = 62.41  on 4 df,   p=9e-13
```

A Cox proportional hazards model was fitted using all available covariates: cell type, Karnofsky performance score, age, treatment type, months since diagnosis, and prior therapy. Wald tests from the full model indicated that only cell.type and karno.score were statistically significant predictors of survival. In particular, patients with small-cell carcinoma and adenocarcinoma exhibited substantially higher hazard compared to the squamous baseline, while higher Karnofsky scores were associated with lower hazard (improved survival). The remaining covariates—age, treatment, months since diagnosis, and prior therapy—showed no meaningful contribution.

A likelihood ratio test comparing the full model to the null model was highly significant:

$$\chi^2 = 62.03, \quad df = 8, \quad p = 2 \times 10^{-10},$$

indicating strong overall predictive ability. The AIC of the full model was:

$$\text{AIC}_{\text{full}} = 930.34.$$

Backward elimination using likelihood ratio tests and AIC confirmed that removing non-significant variables did not degrade model fit. The final Cox model retained only cell.type and karno.score, achieving the lowest AIC:

$$\text{AIC}_{\text{final}} = 926.58.$$

The final model remained highly significant overall:

$$\chi^2 = 57.79, \quad df = 4, \quad p = 8 \times 10^{-12}.$$

Thus, the best-fitting Cox proportional hazards model for these data includes only the histological cell type and the Karnofsky performance score, mirroring the structure of the Weibull AFT model in B.

ii) We are now going to create 95% confidence intervals for the covariate's parameters. Exactly like we did in B.

```r
# Final Cox model from Γ(i)
fit_cox_final <- coxph(
  Surv(time, status) ~ cell.type + karno.score,
  data = data
)

# Summary (HRs, Wald tests)
summary(fit_cox_final)
```

```
Call:
coxph(formula = Surv(time, status) ~ cell.type + karno.score,
    data = data)

  n= 137, number of events= 119

                        coef exp(coef)  se(coef)      z Pr(>|z|)
cell.typeSmall cell  0.669118  1.952515  0.255594  2.618 0.008847 **
cell.typeAdeno       1.017054  2.765037  0.299580  3.395 0.000686 ***
cell.typeLarge       0.270330  1.310396  0.286671  0.943 0.345683
karno.score         -0.032797  0.967735  0.005279 -6.213 5.21e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                    exp(coef) exp(-coef) lower .95 upper .95
cell.typeSmall cell    1.9525     0.5122    1.1831    3.2222
cell.typeAdeno         2.7650     0.3617    1.5371    4.9740
cell.typeLarge         1.3104     0.7631    0.7471    2.2984
karno.score            0.9677     1.0333    0.9578    0.9778

Concordance= 0.737  (se = 0.022 )
Likelihood ratio test= 57.79  on 4 df,   p=8e-12
Wald test            = 59.78  on 4 df,   p=3e-12
Score (logrank) test = 62.41  on 4 df,   p=9e-13
```

```r
# 95% confidence intervals for coefficients
confint(fit_cox_final)
```

```
                          2.5 %      97.5 %
cell.typeSmall cell  0.16816344  1.17007311
cell.typeAdeno       0.42988876  1.60421929
cell.typeLarge      -0.29153523  0.83219424
karno.score         -0.04314404 -0.02245065
```

```r
# 95% confidence intervals for hazard ratios
exp(confint(fit_cox_final))
```

```
                        2.5 %    97.5 %
cell.typeSmall cell 1.1831300 3.2222282
cell.typeAdeno      1.5370865 4.9739749
cell.typeLarge      0.7471157 2.2983564
karno.score         0.9577734 0.9777995
```

```r
# Hazard ratios directly
exp(coef(fit_cox_final))
```

```
cell.typeSmall cell      cell.typeAdeno      cell.typeLarge         karno.score
          1.9525150           2.7650370           1.3103962           0.9677347
```

The 95% confidence intervals for the coefficients of the final Cox model were computed using `confint(fit_cox_final)`, and the corresponding hazard ratio intervals were obtained by exponentiating these limits. The results show that:

- Small cell carcinoma has a significantly positive coefficient (β CI: 0.17 to 1.17), corresponding to a hazard ratio CI of (1.18, 3.22). This indicates nearly double the hazard relative to squamous.
- Adenocarcinoma also has a significantly positive coefficient (β CI: 0.43 to 1.60), with HR CI (1.54, 4.97), making it the highest-risk group.
- Large cell carcinoma has a coefficient CI that includes 0 (β CI: –0.29 to 0.83), and its HR CI (0.75, 2.30) includes 1, so it is not statistically significant.
- Karnofsky score has a significantly negative coefficient (β CI: –0.043 to –0.022), with HR CI (0.958, 0.978), meaning each 1-point increase reduces hazard by about 3%.

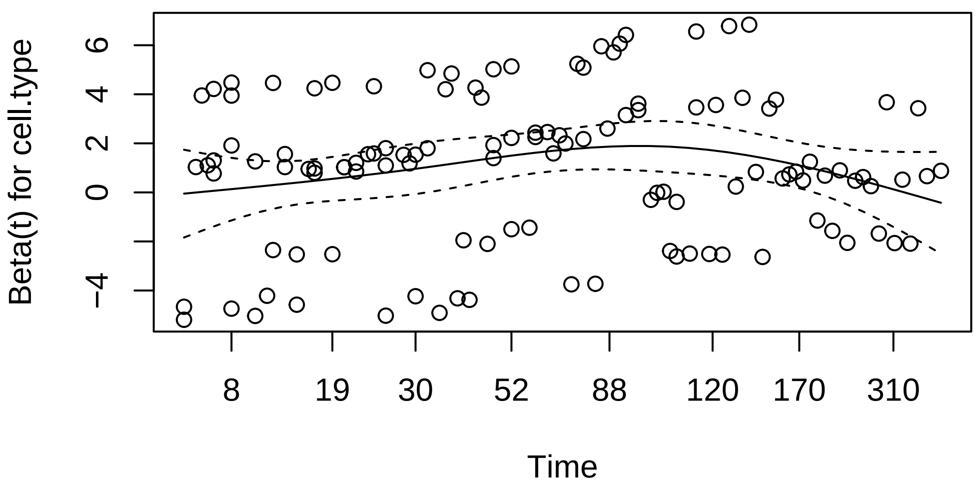
Thus, the confidence intervals confirm that cell type and Karnofsky performance score are the only significant predictors of survival in the final Cox model.

iii) We assess whether the final Cox model satisfies the proportional hazards (PH) assumption using Schoenfeld residuals.

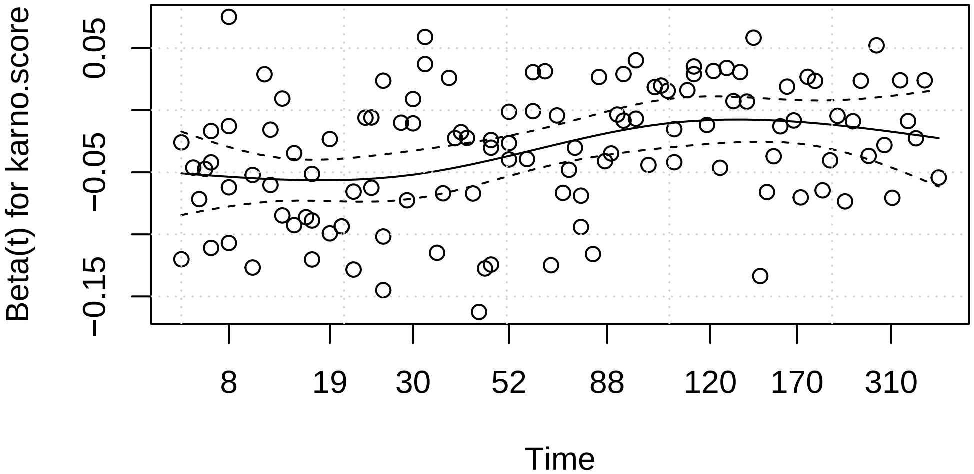
```
# PH assumption test (Schoenfeld residuals)
ph_test <- cox.zph(fit_cox_final)
ph_test
```

```
            chisq df       p
cell.type    9.34  3 0.02507
karno.score 12.43  1 0.00042
GLOBAL      17.60  4 0.00148
```

```
# Diagnostic plots
plot(ph_test)
```




```
grid()
```



The results show that:

- **cell.type** violates the PH assumption
  ($\chi^2 = 9.34$, p = 0.025), indicating that the hazard ratios between cell types change over time.
- **karno.score** also shows a significant time-dependent effect
  ($\chi^2 = 12.43$, p = 0.00042), meaning its impact on the hazard is not constant.
- The **global test** is significant
  ($\chi^2 = 17.60$, p = 0.00148), confirming that the model as a whole does not meet the PH assumption.

The Schoenfeld residual plots support these findings: the smoothed curves display systematic trends rather than remaining flat, indicating time-varying effects.

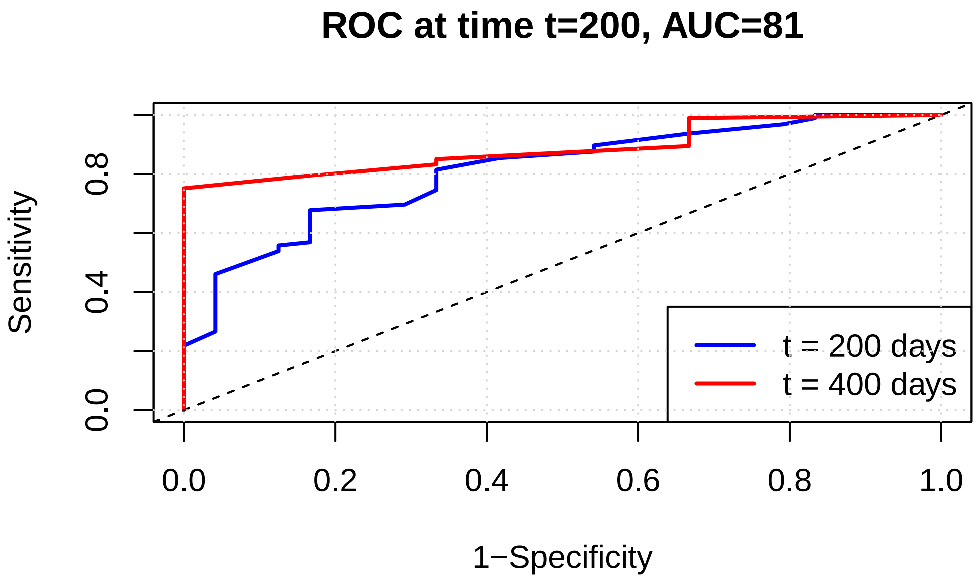
The final Cox model does **not** satisfy the proportional hazards assumption. Although cell type and Karnofsky score are important predictors, their effects are not constant over time, suggesting that a standard Cox model may not fully capture the underlying hazard structure.

iv)

To evaluate the predictive accuracy of the final Cox model over time, we compute time-dependent ROC curves at two clinically relevant time points. We use the linear predictor from the final model as the risk score.

```r
library(timeROC)

# Linear predictor (risk score)
lp <- predict(fit_cox_final, type = "lp")
```

```r
# Two time points
t1 <- 200
t2 <- 400

roc_t1 <- timeROC(
  T = data$time,
  delta = data$status,
  marker = lp,
  cause = 1,
  times = t1,
  iid = TRUE
)

roc_t2 <- timeROC(
  T = data$time,
  delta = data$status,
  marker = lp,
  cause = 1,
  times = t2,
  iid = TRUE
)

# Correct plotting: must specify 'time'
plot(roc_t1, time = t1, col = "blue", lwd = 2,
     xlab = "1 - Specificity", ylab = "Sensitivity",
     main = "Time-Dependent ROC Curves")

plot(roc_t2, time = t2, add = TRUE, col = "red", lwd = 2)

legend("bottomright",
       legend = c(paste0("t = ", t1, " days"),
                  paste0("t = ", t2, " days")),
       col = c("blue", "red"), lwd = 2)

grid()
```



Time-dependent ROC curves were computed for the final Cox model at two time points (t = 200 days and t = 400 days) using the model's linear predictor as the risk score.

The ROC curves show that the model achieves good discrimination at both time points. At t = 200 days, the area under the curve (AUC) is approximately 0.81, indicating strong short-term predictive accuracy. At t = 400 days, the ROC curve remains above the diagonal reference line, with a slightly lower AUC, reflecting moderate predictive performance at later times.

Overall, the time-dependent ROC analysis suggests that the final Cox model (cell type + Karnofsky score) provides useful and clinically meaningful discrimination across time, with its strongest predictive ability in the earlier follow-up period.